Design Guide for Abu Dhabi

Minimum design standards for mechanical services



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Design Guide for Abu Dhabi:

Minimum design standards for mechanical services

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Note from the publisher

This publication is primarily intended to provide guidance to those responsible for the design, installation, commissioning, operation and maintenance of building services. It is not intended to be exhaustive or definitive and it will be necessary for users of guidance to exercise their own professional judgement when deciding whether to abide by or depart from it.

Any commercial products depicted, mentioned or described within this publication are included for the purposes of illustration only and their inclusion does not constitute endorsement or recommendation by the Institution.

This document aims at establishing the minimum design standards for the design of mechanical services for buildings in Abu Dhabi. It does not relieve the designer of responsibility for accurately determining design arrangements and/or complying with the local authority requirements and/or relevant international codes or regulatory requirements. It should be used for initial design guidance, and provides minimum design requirements for specific design issues.

This guide has been produced in good faith by the CIBSE UAE Technical Committee to promote best engineering practices in Abu Dhabi, without legal or financial liability of CIBSE UAE or the authors, in order to assist engineers coming from elsewhere in the world to work in Abu Dhabi, UAE. It should be read alongside existing CIBSE guidance relevant to the region, including *Buildings for Extreme Environments: Arid*.

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1 Design codes, standards and guidance

The design, specification and installation of mechanical services should be in accordance with good engineering practice and should conform to local codes, standards and guidelines, and follow applicable international codes, standards and guidelines where indicated.

1.1 Local codes, standards and guidance

Source/reference	Code, regulation, guideline reference
Department of Municipal Affairs (DMA)	*Abu Dhabi International Mechanical Code* (ADDMA, 2013a)
Department of Municipal Affairs (DMA)	*Abu Dhabi International Energy Conservation Code* (ADDMA, 2013b)
UAE and Abu Dhabi laws and regulations	Circulars and directions of Local Authorities having Jurisdiction
UAE and Abu Dhabi laws and regulations	*UAE Fire and Life Safety Code of Practice* (ADCD, 2018)
Emirates Authority for Standardization and Metrology (ESMA)	UAE.S 5010-1: *Labeling — Energy efficiency label for electrical appliances*; *Part 1: Household air conditioners* (ESMA, 2019)
Abu Dhabi Civil Defense	Circulars
Health Authority Abu Dhabi	HAAD regulations
Abu Dhabi Urban Planning Council	Estidama Guidelines

1.2 International codes, standards and guidance

Source	Code, regulation, guideline reference
American Society of Heating, Refrigerating and Air-conditioning Engineers (ASHRAE)	ASHRAE Standards and guidelines
	Standard 15: *Safety Standard for Refrigeration Systems* (ASHRAE, 2019a)
	Standard 55: *Thermal environmental conditions for human occupancy* (ASHRAE, 2020a)
	Standards 62.1/62.2: *Ventilation for acceptable indoor air quality* (ASHRAE, 2019b, 2019c)
	Standard 90.1/2: *Energy standard for buildings* (ASHRAE, 2007, 2019d, 2018a)
	Standard 169: *Climatic Data for Building Design Standards* (ASHRAE, 2020b)
	Standard 170: *Ventilation for health care facilities* (ASHRAE, 2017a)
	Standard 183: *Peak Cooling and Heating Load Calculations in Buildings Except Low-Rise Residential Buildings* (ASHRAE, 2020c)
	Standard 202: *Commissioning Process for Buildings and Systems* (ASHRAE, 2018b)
National Fire Protection Association (NFPA)	
Chartered Institution of Building Services Engineers	CIBSE Commissioning Codes: • Code A: Air distribution systems (CIBSE, 2006) • Code B: *Boilers* (CIBSE, 2002a) • Code C: *Automatic controls* (CIBSE, 2001) • Code R: *Refrigeration systems* (CIBSE, 2002b) • Code W: *Water distribution systems* (CIBSE, 2010)

Table continues

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Source	Code, regulation, guideline reference
Sheet Metal Air Conditioning Contractors National Association (SMACNA)	
Air-Conditioning, Heating and Refrigeration Institute (AHRI)	
American Society of Mechanical Engineers (ASME)	
American Society for Testing and Materials (ASTM)	
American Society for Health Care Engineering (ASHE)	Health facility commissioning
Underwriter's Laboratories, Inc. (UL)	

2 Climate zone and external conditions

In accordance with Section 301 Table 301.3(2) of the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b), the Emirate of Abu Dhabi shall be Zone 1. As per ASHRAE Standard 169: *Climatic Data for Building Design Standards* (ASHRAE, 2020b) Abu Dhabi is in climate zone OB.

2.1 Minimum building envelope requirements

The designer shall follow project specific envelope values established in coordination with relevant disciplines.

The building projects shall comply with the requirements in Sections 502, 503, 504 and 505 and 506 of the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b). As an alternative the building project shall exceed by at least 25% the requirements of ANSI/ASHRAE/IESNA 90.1-2007, section 11 (ASHRAE, 2007).

Table 2.1 is an excerpt of Tables 502.1.2 and 502.3 of the *Abu Dhabi International Energy Conservation Code*.

Table 2.1 Maximum *U*-values and solar heat gain coefficients (source: *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b))

Building fabric details	Maximum *U*-values (W/m^2·K) and solar heat gain/shading coefficients
External wall mass above grade	*U*-value: 0.34
Windows (30% maximum of above-grade wall)	*U*-value: 2.2 Solar heat gain coefficient for stated projection factor* (*PF*): • $PF < 0.25$: 0.25 • $0.25 \leq PF < 0.5$: 0.33 • $PF \geq 0.5$: 0.4
Skylights (3% maximum)	*U*-value: 4.2 Shading coefficient: 0.35
Roof	*U*-value: 0.22
Slab on grade floors	Not required

* Projection factor (*PF*) = *A*/*B* where *A* is the distance measured horizontally from the furthest continuous extremity of any overhang, eave, or permanently attached shading device to the vertical surface of the glazing and *B* is the distance measured vertically from the bottom of the glazing to the underside of the overhang, eave, or permanently attached shading device.

Where different windows or glass doors have different *PF* values, they shall each be evaluated separately, or an area-weighted *PF* value shall be calculated and used for all windows and glass doors.

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2.2 External design conditions

External design conditions for space cooling and fresh air equipment shall be in accordance with client directive and/or ASHRAE Handbook: *Fundamentals* (2017a).

Considering the hot and humid nature of the climate of the Emirate of Abu Dhabi, it is recommended to use annual dehumidification conditions to size fresh air equipment coils.

The Emirate of Abu Dhabi is divided in three regions: Abu Dhabi, Al Ain and Al Dhafra. Each region presents different external conditions. Annual space cooling (0.4%) and space heating conditions (99.6%) for the three regions are shown in Tables 2.2, 2.3 and 2.4 respectively.

Table 2.2 External design conditions: Abu Dhabi

Parameter	Value	Data source
Dry bulb design temperature	44.9 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'
Mean coincident wet bulb design temperature	23.0 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'
Winter dry bulb temperature	11.8 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'

Table 2.3 External design conditions: Al Ain

Parameter	Value	Data source
Dry bulb design temperature	46.0 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'
Mean coincident wet bulb design temperature	22.8 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'
Winter dry bulb temperature	10.9 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'

Table 2.3 External design conditions: Al Dhafa

Parameter	Value	Data source
Dry bulb design temperature	46.0 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'
Mean coincident wet bulb design temperature	23.1 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'
Winter dry bulb temperature	10.9 °C	ASHRAE Handbook: *Fundamentals*, chapter F14, 'Climatic design information'

3 Indoor design conditions

The interior design temperatures used for heating and cooling load calculations shall be a maximum of 22 °C for heating and a minimum of 24 °C for cooling with a relative humidity of 50 ± 5% as per Section 302.1 of the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b).

Detail space by space conditions should follow project specific requirements in coordination with relevant stakeholders.

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4 Cooling and heating load calculations

Design loads shall be determined in accordance with the procedures described in the ASHRAE/ACCA Standard 183: *Peak Cooling and Heating Load Calculations in Buildings Except Low-Rise Residential Buildings* (ASHRAE, 2020c) The design loads shall account for the building envelope, equipment, lighting, ventilation and occupancy loads based on the project design.

ASHRAE Handbook: *Fundamentals* (ASHRAE, 2017b) can be referred to for further information on cooling load calculations.

Heating and cooling loads shall be adjusted to account for load reductions that are achieved when energy recovery systems are utilized in the HVAC system in accordance with ASHRAE Handbook: *HVAC Systems and Equipment* (ASHRAE, 2020d). Alternatively, design loads shall be determined by an approved equivalent computation procedure, using the design parameters specified in Chapter 3 of the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b).

4.1 Internal loads

People density should be based on the architectural furniture arrangement; where this is not available it should then be based on *Abu Dhabi International Mechanical Code* (ADDMA, 2014a), Table 403.3, or latest ASHRAE Standards 62.1 and 62.2 (ASHRAE, 2019b, 2020c). The associated sensible and latent loads should be based on ASHRAE Handbook: *Fundamentals* (ASHRAE, 2017b).

Equipment heat gains should be based on actual equipment heat dissipation of project specific components. In the absence of such information, ASHRAE *Fundamentals* should be used.

4.1.1 Lighting power density: interior

Lighting power density should be based on *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b) requirements, as shown in Table 4.1.

Design power density consideration should be in line with project-specific lighting design and sustainability requirements. For new buildings, the average lighting power density for the interior connected lighting load must not exceed the values from the A*bu Dhabi International Energy Conservation Code*, Table 505.5.2, for the whole building and space by space. Whole building lighting power densities are summarized in Table 4.1 (values have been rounded up in converting from W/ft^2 to W/m^2).

Table 4.1 Whole building lighting power densities

Building type	Maximum average (W/m^2) across total building area
Automotive facility	10
Convention centre	13
Courthouse	13
Dining:	
— bar lounge/leisure	14
— cafeteria/fast food	15
— family	17
Dormitory	11
Exercise centre	11

Table continues

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Table 4.1 Whole building lighting power densities (*continued*)

Building type	Maximum average (W/m²) across total building area
Gymnasium	12
Healthcare clinic	11
Hotel	11
Library	14
Manufacturing facility	14
Motel	11
Motion picture theatre	13
Multi-family	8
Museum	12
Office	10
Parking garage	4
Parking garage	4
Penitentiary	11
Performing arts theatre	17
Police station	11
Fire stations	9
Post office	12
Religious buildings	14
Retail:	14
— specialty	17
— supermarket	14
School/university	13
Town Hall	12
Transportation	11
Sports arena	12
Warehouse	7
Workshop	15

Lighting power density values for the building types not listed in Table 4.1 should not exceed the values indicated in the latest edition of ASHRAE Standards 90.1 and 90.2 (ASHRAE, 2019d, 2018a) or equivalent as approved by the Authority. Consultants seeking a Pearl Rating (Estidama) should follow reduced lighting power density as described in the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b), Table 506.3.

5 Infiltration rates

The design should consider the infiltration rates indicated in the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b), Section 502.4, to comply with the building air barrier requirements. Assemblies of materials and components shall have an average leakage not to exceed 0.2 litre/s·m² at ΔP of 75 Pa. A maximum leakage rate of 2.0 litre/s·m² of building envelope at ΔP of 75 Pa shall not be exceeded for the building test. When applying an infiltration rate in the cooling load software the ΔP used to recalculate the rate has to be changed to a more representative pressure that the building will be subjected to during operation. The recommended range for a mid-rise building is 5 to 12 Pa. This will highly depend on the height of the building and wind conditions. The infiltration from doors to the outside that are expected to be used frequently should be taken into account in the infiltration calculation.

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6 Ventilation rates

The ventilation rates should generally be based on the occupancy density and in accordance with the *Abu Dhabi International Mechanical Code* (ADDMA, 2014a), Table 403.3 'Minimum Ventilation Rates', and ASHRAE Standards 62.1, 62.2 and 170 (ASHRAE, 2019b, 2019c, 2017a).

7 Diversity and safety factor

The design should account for diversity profiles for people, lighting, equipment, etc., and these should be considered in the air conditioning plant calculations.

Sensible and latent safety factors of 10% and 5% respectively can be applied based on good engineering practice in calculating space by space heat load and sizing air conditioning equipment.

8 Building pressurization

The air conditioning and ventilation systems should be designed such that the building will be maintained at a positive pressure. Generally, the building should be pressurized with respect to the outside by approximately 10%; the outside air quantity should be 10% higher than the exhaust air quantity. This will assist in limiting infiltration of unconditioned ambient air.

9 Air movement

The air movement within the occupied zone should be designed to be in accordance with ASHRAE Standard 55 (ASHRAE, 2020a) to prevent draughts and the associated feelings of discomfort and is recommended to be between the limits of 0.20 m/s when occupants are neutral to slightly cool (operative temperature of 23 °C) and 0.3 m/s in warmer conditions and as prescribed in the above standard.

10 Noise and vibration control

Building services should be designed and installed in accordance with the advice of the project acoustic consultant. In absence of the such advice, ASHRAE Handbook: *Fundamentals* (ASHRAE, 2017b) section F08, 'Sound and vibration', or CIBSE Guide B4: *Noise and vibration control for building services systems* (2016) and CIBSE Guide A: *Environmental design* (2015) should be followed. The acoustic performance relating to internal noise criteria from various sources should meet project requirements.

11 Energy conservation

Systems should be designed to minimize energy use in line with and the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b), chapters 4 and 5.

Energy conservation methods should be considered for all mechanical systems. Fans and pumps should be provided with variable speed drives where possible to achieve energy saving targets.

For air handling units (AHUs) that comprise supply and extract in one unit, consideration should be given to energy recovery systems. Thermal wheels, sensible wheels, plate heat exchangers and/or heat pipes should be considered for the recovery of sensible and latent

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loads and the relative merits, including a quantitative assessment of efficiencies, and should be in accordance with ASHRAE Handbook: *HVAC Systems and Equipment* (2020b).

Demand control ventilation (DCV) is to be applied in all areas that have a maximum occupancy density of 40 people per 93 m^2, for spaces larger than 50 m^2 as highlighted in the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b), Section 503.2.5.1. Demand control ventilation can include the installation of carbon dioxide (CO_2) sensors, which should control and minimize the amount of fresh air supplied to the space while maintaining the required CO_2 levels. During pandemics, especially where transmission is suspected to be airborne, it is recommended to disable DCV and keep the outside air to the maximum.

Effective energy management requires close and consistent control of all energy-consuming systems and components applicable to the project.

12 Refrigerants

All refrigerants should comply with the requirements of the *Abu Dhabi International Mechanical Code* (ADDMA, 2014a), section 1103 and Table 1103.1.

Ozone-depleting substances are prohibited for new installations and projects seeking a Pearl Rating (Estidama) shall consider using refrigerants with zero ozone depletion potential and low global warming potential, as highlighted in the *Abu Dhabi International Mechanical Code*.

13 Cooling towers

Cooling towers should be selected to achieve the minimum energy performance requirements of the latest version of ASHRAE 90.1 (ASHRAE, 2019d). All cooling towers should be CTI-certified* and should be selected at the evaporation conditions of the respective regions, unless otherwise agreed with the client.

14 Chiller coefficient of performance (CoP)

Chiller CoP should be in accordance with the *Abu Dhabi International Energy Conservation Code* (ADDMA, 2014b), Tables 503.2.3(7), 506.2(4) and 506.2(5).

15 Hydraulic pressures and temperature

Water pipework and ductwork system operating/working and test pressures and temperatures should be determined in accordance with the design parameters set by the mode of operation of that particular system and in full compliance with the appropriate specification, ASHRAE guidelines, local code of practice or CIBSE Guide and all relevant regulations.

16 Fire and life safety systems

The ventilation systems associated with fire and life safety (FLS) should be informed by the FLS strategy for the building in accordance with the *UAE Fire and Life Safety Code of Practice* (ADCD, 2018) and Abu Dhabi Civil Defense.

* Cooling Technology Institute (https://www.coolingtechnology.org/cti-certification)

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For basement car parking, the extract rate should be 6 air changes per hour (ACH) for normal mode and 10 ACH for smoke mode, with the necessary air make-up provision within the car parking area. The system should be designed in accordance with the *UAE Fire and Life Safety Code of Practice* (ADCD, 2018) chapter 10, 'Smoke control and smoke management systems'.

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